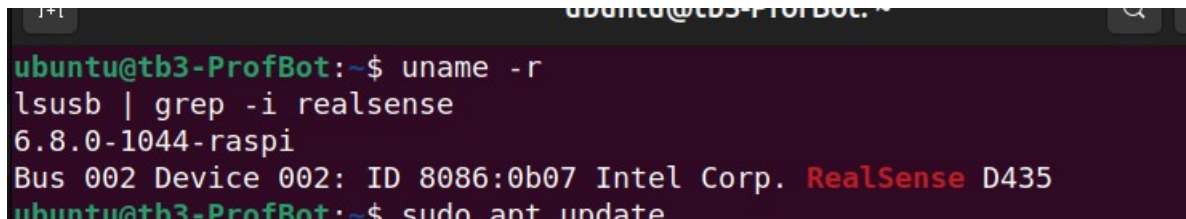


Intel Realsense auf dem TB3 – Jazzy

Kamera per USB (blau) anstecken und kontrollieren ob Sie erkannt wird.
Dazu den TB3 starten und per SSH mit dem Remote-PC verbinden.

```
uname -r  
lsusb | grep -i realsense
```



```
ubuntu@tb3-ProfBot:~$ uname -r  
6.8.0-1044-raspi  
ubuntu@tb3-ProfBot:~$ lsusb | grep -i realsense  
Bus 002 Device 002: ID 8086:0b07 Intel Corp. RealSense D435  
ubuntu@tb3-ProfBot:~$ sudo apt update
```

Zielzustand

Am Ende läuft auf dem TurtleBot3:

- realsense2_camera Node (ROS 2 Jazzy)
- Topics:
 - /camera/depth/image_rect_raw
 - /camera/depth/color/points
- TF: camera_link → base_link

System vorbereiten (einmalig) kann länger dauern

```
sudo apt update  
sudo apt update  
sudo apt upgrade -y  
sudo apt install -y \  
  build-essential \  
  cmake \  
  git \  
  pkg-config \  
  libusb-1.0-0-dev \  
  libssl-dev \  
  libglfw3-dev \  
  libgtk-3-dev \  
  libgl1-mesa-dev \  
  python3-rosdep \  
  python3-colcon-common-extensions
```

ROS Umgebung

```
source /opt/ros/jazzy/setup.bash  
sudo rosdep init || true
```

```
rosdep update
```

Librealsense - Build from Source

```
cd ~
git clone https://github.com/IntelRealSense/librealsense.git
cd librealsense
```

Udev-Regeln installieren (sehr wichtig!)

```
sudo cp config/99-realsense-libusb.rules /etc/udev/rules.d/
sudo udevadm control --reload-rules
sudo udevadm trigger
```

```
sudo usermod -aG plugdev $USER
logout
```

=> Neu Einloggen

Build (minimal, ohne GUI-Zeug)

```
cd ~/librealsense
mkdir -p build
cd build

cmake .. \
  -DBUILD_EXAMPLES=false \
  -DBUILD_GRAPHICAL_EXAMPLES=false \
  -DBUILD_WITH_TM2=false \
  -DFORCE_RSUSB_BACKEND=true

make -j2
sudo make install
sudo ldconfig
```

Test (ohne ROS)

```
realsense-viewer
```

Mit ROS

```
cd ~/turtlebot3_ws/src
git clone https://github.com/IntelRealSense/realsense-ros.git
rosdep install --from-paths src --ignore-src -r -y

cd ..
colcon build --symlink-install --parallel-workers 1
source ~/.bashrc
```

--parallel-workers 1: Dieser Parameter zwingt colcon, nur **ein Paket gleichzeitig** zu bauen.

- **Vorteil:** Dies ist extrem hilfreich auf Systemen mit **wenig Arbeitsspeicher** (z. B. Raspberry Pi), da gleichzeitige Kompilierungsvorgänge oft den RAM sprengen und zum Einfrieren des Systems führen

RealSense starten (ressourcenschonend für Pi 4)

```
rros2 run realsense2_camera realsense2_camera_node --ros-args \
  -p enable_depth:=true \
  -p enable_color:=false \
  -p enable_infra1:=false \
```

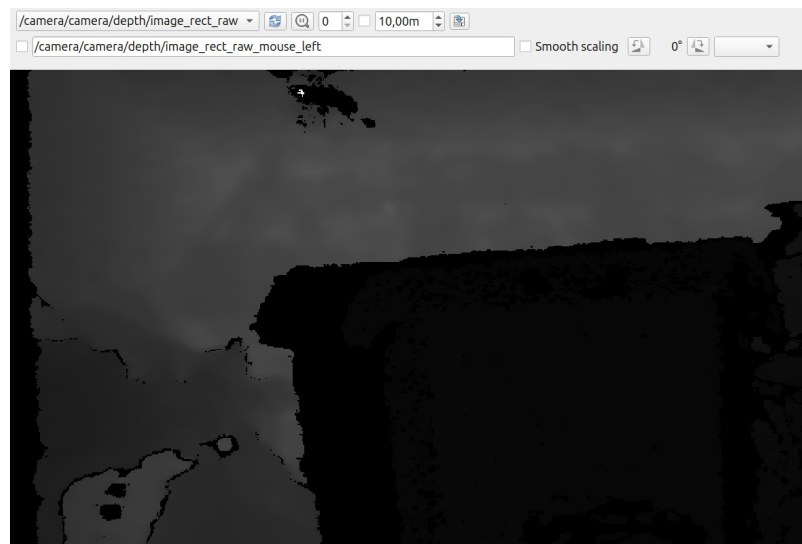
```
-p enable_infra2:=false \  
-p pointcloud.enable:=true \  
e
```

```
ubuntu@tb3-ProfBot: ~/turtlebot3_ws  
oj@HomeOffice:~$ ros2 topic list  
/camera/camera/depth/camera_info  
/camera/camera/depth/image_rect_raw  
/camera/camera/depth/metadata  
/cmd_vel  
/parameter_events  
/rosout  
/tf_static  
oj@HomeOffice:~$
```

```
-p pointcloud.stream_filter:=2 \  
-p depth_module.profile:="640x480x6"
```

```
[INFO] [1768564071.036999085] [camera.camera]: Close Sensor - Done.  
ubuntu@tb3-ProfBot:~/turtlebot3_ws$ ros2 run realsense2_camera realsense2_camera_node --ros-args \  
-p enable_depth:=true \  
-p enable_color:=false \  
-p enable_infra1:=false \  
-p enable_infra2:=false \  
-p pointcloud.enable:=true \  
-p pointcloud.stream_filter:=2 \  
-p depth_module.profile:="640x480x15"  
[INFO] [1768564083.002479042] [camera.camera]: RealSense R0S v4.57.4  
[INFO] [1768564083.002727392] [camera.camera]: Built with LibRealSense v2.57.5  
[INFO] [1768564083.002786169] [camera.camera]: Running with LibRealSense v2.57.5  
16/01 12:48:03.004 ERROR [281472415883328] (context.cpp:40) No valid configuration file found at : /home/u  
[INFO] [1768564083.443169930] [camera.camera]: Device with serial number 239722071217 was found.  
  
[INFO] [1768564083.443408595] [camera.camera]: Device with physical ID 2-2-2 was found.  
[INFO] [1768564083.443480835] [camera.camera]: Device with name Intel RealSense D435 was found.  
[INFO] [1768564083.444125071] [camera.camera]: Device with port number 2-2 was found.  
[INFO] [1768564083.444263274] [camera.camera]: Device USB type: 3.2  
[INFO] [1768564083.444515642] [camera.camera]: getParameters...  
[INFO] [1768564083.455617005] [camera.camera]: JSON file is not provided  
[INFO] [1768564083.455777855] [camera.camera]: Device Name: Intel RealSense D435  
[INFO] [1768564083.455839484] [camera.camera]: Device Serial No: 239722071217  
[INFO] [1768564083.455883114] [camera.camera]: Device physical port: 2-2-2  
[INFO] [1768564083.455926928] [camera.camera]: Device FW version: 5.17.0.10  
[INFO] [1768564083.455964687] [camera.camera]: Device Product ID: 0x0B07  
[INFO] [1768564083.456004539] [camera.camera]: Sync Mode: Off  
[INFO] [1768564085.154358956] [camera.camera]: Stopping Sensor: Depth Module  
[INFO] [1768564085.244650806] [camera.camera]: Starting Sensor: Depth Module  
[INFO] [1768564085.350045730] [camera.camera]: Open profile: stream_type: Depth(0), Format: Z16, Width: 848  
16/01 12:48:05.354 WARNING [281472378134592] (messenger-libusb.cpp:42) control_transfer returned error, in  
number: 11  
16/01 12:48:05.369 WARNING [281472378134592] (messenger-libusb.cpp:42) control_transfer returned error, in  
number: 11  
[INFO] [1768564085.414343347] [camera.camera]: RealSense Node Is Up!
```

```
ros2 run realsense2_camera realsense2_camera_node --ros-args -p enable_depth:=true -p  
enable_color:=false -p enable_infra1:=false -p enable_infra2:=false -p pointcloud.enable:=true  
-p pointcloud.stream_filter:=2 -p depth_module.profile:="640x480x6"
```



```
min: 0.176s max: 10.283s std dev: 0.44496s window: 513  
^Coj@HomeOffice:~$ ros2 topic hz /camera/camera/depth/image_rect_raw  
average rate: 5.107  
min: 0.194s max: 0.200s std dev: 0.00185s window: 7  
average rate: 5.113  
min: 0.194s max: 0.200s std dev: 0.00148s window: 13  
average rate: 5.114  
min: 0.194s max: 0.200s std dev: 0.00129s window: 19  
average rate: 5.119  
min: 0.194s max: 0.200s std dev: 0.00120s window: 25  
average rate: 5.121
```

Kamera und TB3-Launch laufen parrallel:

```
oj@HomeOffice:~$ ssh ubuntu@192.168.178.39
ubuntu@192.168.178.39's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1044-raspi aarch64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Fri Jan 16 01:02:26 PM CET 2026

```
System load:          1.48
Usage of /:            75.8% of 14.07GB
Memory usage:         22%
Swap usage:           3%
Temperature:          65.7 C
Processes:            170
Users logged in:      1
IPv4 address for wlan0: 192.168.178.39
IPv6 address for wlan0: fdb0:460:c057:0:e65f:1ff:feb7:7576
IPv6 address for wlan0: 2a00:6020:a3bb:7d00:e65f:1ff:feb7:7576
```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

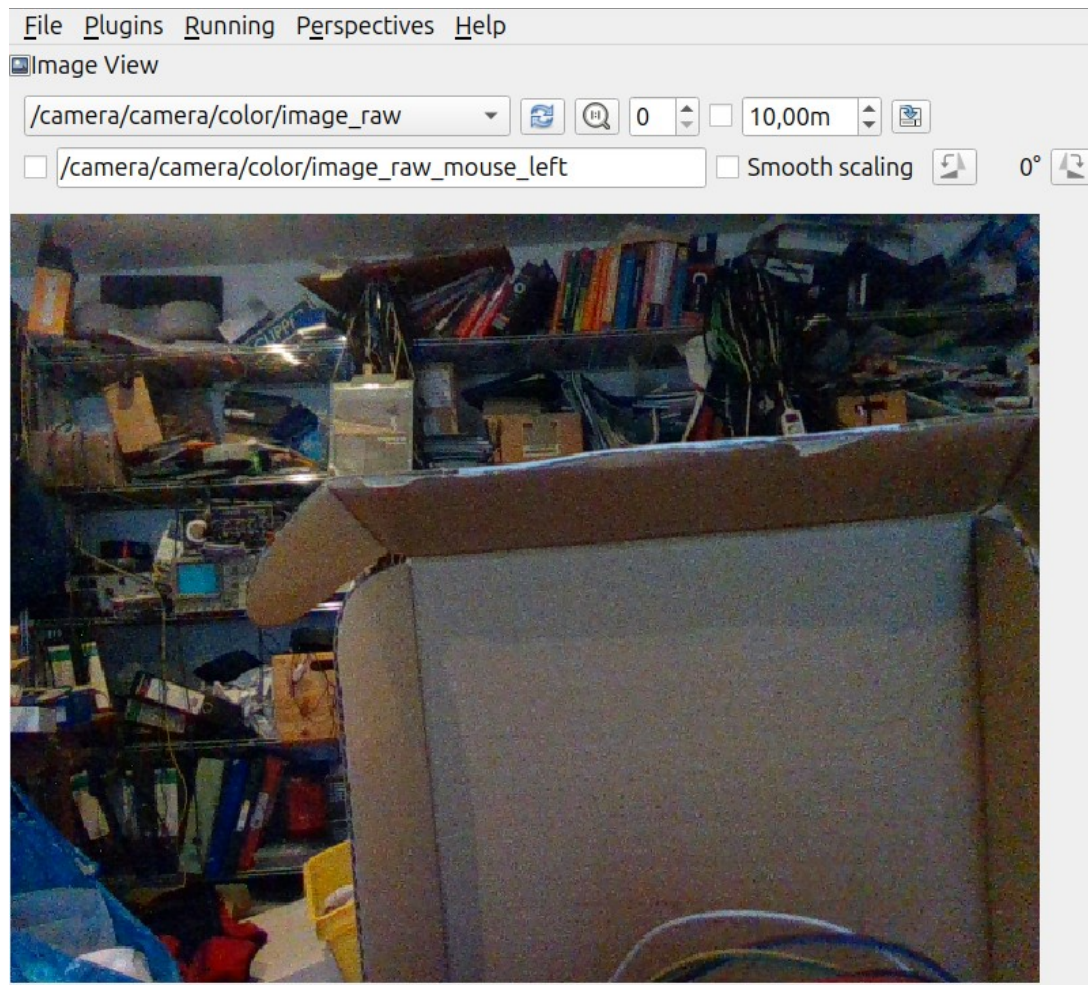
Last login: Fri Jan 16 13:01:24 2026 from 192.168.178.22

```
ubuntu@tb3-ProfBot:~$ top
```

```
top - 13:03:07 up 4:02, 3 users, load average: 1.66, 1.49, 2.12
Tasks: 168 total, 1 running, 167 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.7 us, 13.8 sy, 0.0 ni, 82.3 id, 0.0 wa, 0.0 hi, 0.3 s
MiB Mem : 1839.3 total, 1096.4 free, 458.9 used, 466.8 buff
MiB Swap: 2048.0 total, 1983.8 free, 64.2 used. 1380.4 avai
```

Nur RGB-Kamera

```
ros2 run realsense2_camera realsense2_camera_node --ros-args -p enable_depth:=false -p
enable_color:=true -p enable_infra1:=false -p enable_infra2:=false -p pointcloud.enable:=false
-p pointcloud.stream_filter:=2 -p depth_module.profile:="640x480x6"
```



TF: Kamera an TurtleBot3 anbinden

```
ros2 run tf2_ros static_transform_publisher \  
  0.05 0.0 0.20 \  
  0 0 0 \  
  base_link camera_link
```

Nav2-Integration (empfohlen: VoxelLayer)

```
voxel_layer:  
  plugin: "nav2_costmap_2d::VoxelLayer"  
  enabled: true  
  publish_voxel_map: false  
  origin_z: 0.0  
  z_resolution: 0.05  
  z_voxels: 16  
  max_obstacle_height: 1.2  
  mark_threshold: 0  
  observation_sources: realsense  
  
realsense:  
  topic: /camera/depth/color/points  
  data_type: "PointCloud2"  
  marking: true  
  clearing: true  
  obstacle_min_range: 0.3  
  obstacle_max_range: 2.0
```



```
raytrace_min_range: 0.3  
raytrace_max_range: 3.0
```

Sehr wichtige Praxis-Tipps (Pi 4)

USB

- Kamera **direkt** an USB3-Port (blau)
- Kein passiver Hub
- Wenn instabil → aktiver USB3-Hub

Performance

- 15 FPS reichen für Nav2 völlig
- `pointcloud.enable:=true` nur, wenn wirklich benötigt
- Alternativ: **Depth** → **LaserScan** (deutlich weniger CPU)

Ja – für die **D435** gibt es unter ROS 2 den etablierten „Treiber-Stack“:

1. **librealsense** (Intel RealSense SDK, stellt Kamera/USB/Depth bereit)

realsense-ros (ROS 2 Wrapper, Node `realsense2_camera`, publiziert `/image`, `/camera_info`, `/points`, `/imu` usw.)

Aber: Auf Raspberry Pi 4 (arm64) + Ubuntu 24.04 ist es typischerweise kein “apt install und fertig”

Für `realsense2_camera` gab es historisch (und in der Praxis oft noch) **keine arm64-Binary-Pakete** im ROS-Repository; viele Nutzer bauen daher den Wrapper aus Source.

Auch `librealsense` wird auf Raspberry Pi häufig aus Source gebaut (oder mit den Intel-Skripten), inkl. `udev`-Regeln.

Empfohlener Weg auf TB3 (Ubuntu 24.04 Server, Pi 4)

1) librealsense installieren (Source / Build-Guide)

- Folgen Sie dem Intel Build-Guide für Ubuntu/Linux (Dependencies, Build, `udev` rules).

Prüfen Sie Kernel-/Plattform-Support: `librealsense` nennt Ubuntu 24.04 als supported Plattform (Kernelabhängigkeiten beachten).

Wichtig: **udev-Regeln** aktivieren, sonst fehlen Ihnen Rechte auf das USB-Gerät.

2) realsense-ros (ROS2 Wrapper) aus Source im Colcon-Workspace

Auf arm64 ist das meist der realistischste Weg (statt `apt install`).

Grobablauf:

- `~/tb3_ws/src` (oder eigenes `realsense_ws/src`)
 - Repo `realsense-ros` klonen (Branch passend zu Ihrer ROS2-Distro)
 - `rosdep install ...`
 - `colcon build`
 - `source install/setup.bash`
-

Praxis-Hinweise speziell für Pi 4 + D435

Die D435 kann den Pi 4 leicht überfordern, wenn Sie “alles” (RGB + Depth + PointCloud in hoher Auflösung/FPS) aktivieren.

Bewährt:

- **Depth-Auflösung/FPS reduzieren** (z. B. 640×480 @ 15 FPS)
 - **PointCloud nur bei Bedarf** aktivieren (oder Downsampling/Filter)
 - Für Nav2 häufig besser: **Depth** → **2D LaserScan** (z. B. `depthimage_to_laserscan`), statt volle Punktwolke in die Costmap zu schieben (weniger CPU, stabiler)
-

Kompilierproblem gelöst:

Das Problem:

```
[Processing: realsense2_camera]
--- stderr: realsense2_camera
In file included from
/home/ubuntu/turtlebot3_ws/src/realsense-ros/realsense2_camera/include/
ros_sensor.h:22,
                from
/home/ubuntu/turtlebot3_ws/src/realsense-ros/realsense2_camera/src/./include/
base_realsense_node.h:63,
                from
/home/ubuntu/turtlebot3_ws/src/realsense-ros/realsense2_camera/src/
rs_node_setup.cpp:15:
/home/ubuntu/turtlebot3_ws/src/realsense-ros/realsense2_camera/include/
ros_utils.h:36:36: error: 'RS2_STREAM_SAFETY' was not declared in this scope;
did you mean 'RS2_STREAM_ANY'?
   36 |         const stream_index_pair SAFETY{RS2_STREAM_SAFETY, 0};
```

Die Lösung

```
ubuntu@tb3-ProfBot:~/turtlebot3_ws$ pkg-config --modversion realsense2 || true
```



```

ls /usr/local/include/librealsense2/rs.hpp 2>/dev/null && echo "usr/local
librealsense headers vorhanden"
ls /opt/ros/jazzy/include/librealsense2/rs.hpp && echo "ROS librealsense headers
vorhanden"
2.57.5
/usr/local/include/librealsense2/rs.hpp
usr/local librealsense headers vorhanden
/opt/ros/jazzy/include/librealsense2/rs.hpp
ROS librealsense headers vorhanden
ubuntu@tb3-ProfBot:~/turtlebot3_ws$ dpkg -l | grep -E "ros-jazzy-.*realsense|
ros-jazzy-.*librealsense" || true
ii  ros-jazzy-librealsense2                2.56.4-
1noble.20250722.140856          arm64      Library for controlling and
capturing data from the Intel(R) RealSense(TM) D400 devices.
ubuntu@tb3-ProfBot:~/turtlebot3_ws$ sudo apt remove -y ros-jazzy-librealsense2
sudo apt autoremove -y
sudo ldconfig
[sudo] password for ubuntu:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  ros-jazzy-librealsense2
0 upgraded, 0 newly installed, 1 to remove and 2 not upgraded.
After this operation, 32.6 MB disk space will be freed.
(Reading database ... 183521 files and directories currently installed.)
Removing ros-jazzy-librealsense2 (2.56.4-1noble.20250722.140856) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
ubuntu@tb3-ProfBot:~/turtlebot3_ws$ ls
/opt/ros/jazzy/include/librealsense2/rs.hpp && echo "noch da" || echo "ROS
librealsense Header sind weg"
ls: cannot access '/opt/ros/jazzy/include/librealsense2/rs.hpp': No such file or
directory
ROS librealsense Header sind weg
ubuntu@tb3-ProfBot:~/turtlebot3_ws$ cd ~/turtlebot3_ws
rm -rf build/realsense2_camera build/realsense2_camera_msgs \
install/realsense2_camera* install/realsense2_camera_msgs* \
log
ubuntu@tb3-ProfBot:~/turtlebot3_ws$ source /opt/ros/jazzy/setup.bash

export CMAKE_PREFIX_PATH=/usr/local:$CMAKE_PREFIX_PATH
export PKG_CONFIG_PATH=/usr/local/lib/pkgconfig:$PKG_CONFIG_PATH
export LD_LIBRARY_PATH=/usr/local/lib:$LD_LIBRARY_PATH
export realsense2_DIR=/usr/local/lib/cmake/realsense2

colcon build --symlink-install --parallel-workers 1 --cmake-force-configure

```